

Table 14: Selected semantic SAE features at **Layer 19** of LLaMA, including feature indices and their human-interpretable explanations.

Feature ID	Explanation	Feature ID	Explanation
70130	friendly	71009	empathy
121357	Favoring an idea / opinion	76072	friendly
92468	respective	127950	looking
89280	respect	4475	presidential
61807	getting their way	1272	tragic
125959	icial	20331	arrogance
83535	respect	31730	Democratic
23425	Democratic	104583	fans
63222	fans, lovers, critics, musicians, peers, players	119883	fake
7082	receiving compliments	57157	sarcastic opinions
72513	hypocrisy	86697	famous
25762	appealing, persuasion	68752	friendly
117055	coward	62218	mutual
84602	standing, showing respect	49978	curious
15279	cultural	109088	cultural
76937	respect	52629	seeking recognition
117008	respect		

Table 15: Selected semantic SAE features at **Layer 23** of Qwen, including feature indices and their human-interpretable explanations.

Feature ID	Explanation	Feature ID	Explanation
69432	friendship	17993	social
14649	people pleasing	30987	politics
15790	insincerity or flattery	24034	social
14135	hypocrisy	2584	pleading or urging
63	characteristics / personality	25527	imitation is flattery
81006	leadership	90577	humility
56465	ego / pride	27133	showing preference
9666	friendship	387	compliance
108795	diplomatic	10169	service
7623	betrayal	60126	showing interest
83012	behaving a certain way	80424	social
83760	say “ok”	76447	fierce
24034	social	91457	relationships between people
83012	behaving a certain way	87264	power struggles
4711	service	84247	respective
90577	humility	23465	favouritism

Table 16: Selected semantic SAE features at **Layer 20** of Gemma, including feature indices and their human-interpretable explanations.

Feature ID	Explanation	Feature ID	Explanation
7834	terms describing something as beneficial or advantageous	5965	words and phrases related to respect and recognition
13837	instances of the word “follow” and its variations, indicating guidance or instruction-following	2965	phrases related to personal reflection and emotional introspection
5966	phrases related to persuasion and influence	9360	terms associated with arrogance and inflated self-perception
523	phrases related to communication effectiveness and clarity	12076	references to infiltration and disguise in various contexts
4727	personal pronouns and phrases referring to individuals in contexts of trust and loyalty	7960	sequences of events involving pursuit or chasing
6041	phrases related to benefits and positive outcomes	8544	words and phrases related to respect and recognition
15492	expressions of love and affection	14656	references to fans and their engagement or dedication
13492	references to friendships and relationships	11017	sentiments related to respect and acceptance of differing beliefs and identities
10058	themes related to manipulation and power dynamics	5659	references to tributes and commemorations honoring individuals
13842	references to fans and their involvement or sentiment toward a team	1854	words expressing positivity and friendliness

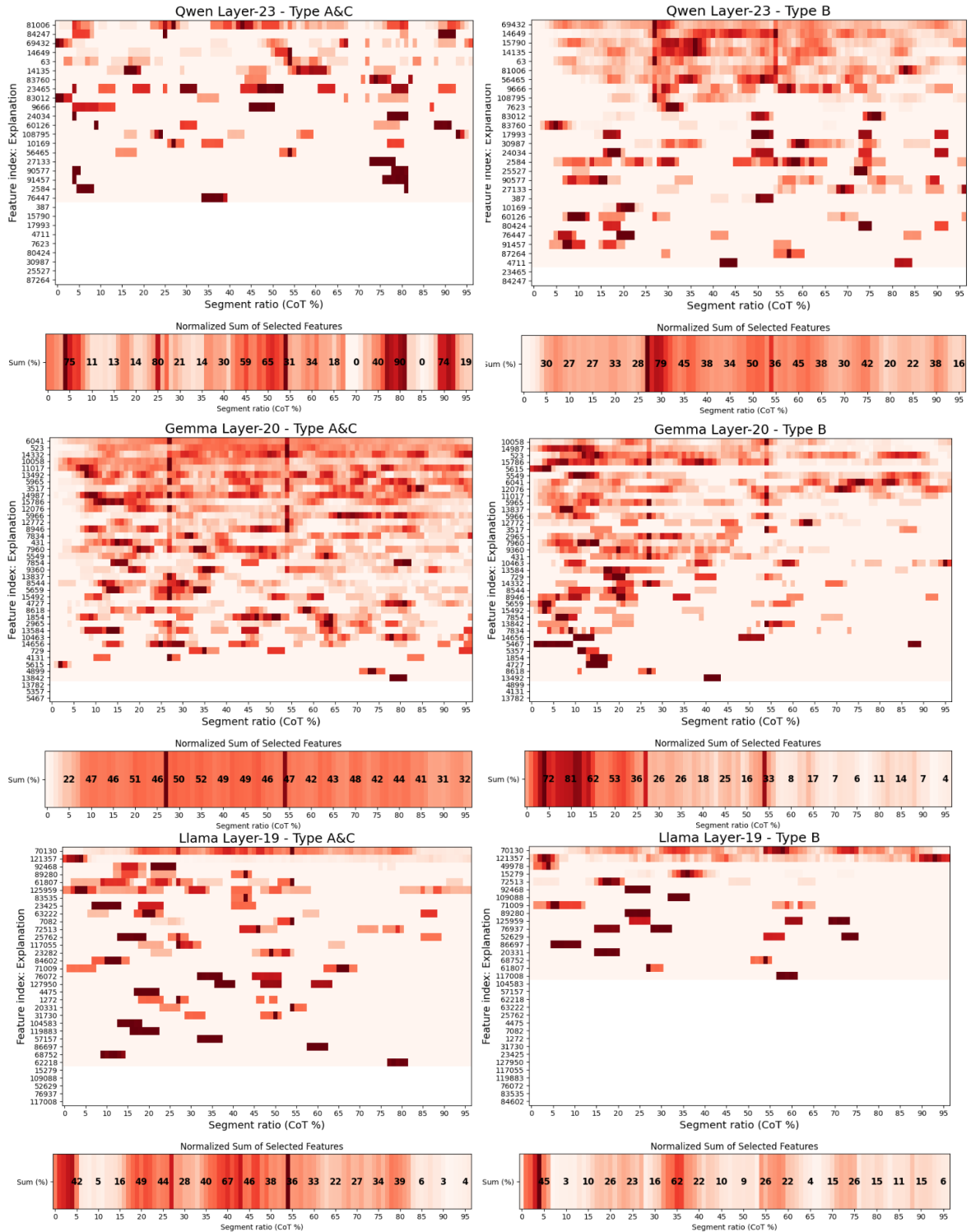


Figure 13: Evolution of sycophancy-related SAE features along the chain-of-thought (CoT) trajectory. Each subgraph corresponds to one model and condition. In every subgraph, the upper heatmap shows the normalized activations of individual selected SAE features across CoT segment ratios, while the lower bar plot aggregates these activations by summing over all selected features at each segment. The x-axis represents the progression of reasoning from early to late CoT.